Software Requirement Specification

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Clarity	1					
Content Quality	2					
Spelling & Grammar	1					
Organization and Formatting	1					
					Total obtained mark	
Comments						

Semester: Fall 2026

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Batch: 42 Section: B2 Course Code: SE331

Course Name: Software Engineering Design Capstone Project

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1. Introduction

This section gives a high-level overview of the project's purpose and goals.

1.1 Problem Statement

Organizations such as schools and universities regularly buy, store and consume goods (for example printer paper, toner, laboratory supplies, cleaning materials, computer accessories and furniture). Their purchases, consumption, prices and expenses are usually kept in spreadsheets, paper ledgers or basic inventory software that is used only for record keeping. The organization therefore holds historical data, but that data does not automatically help it decide what to buy, how much to buy and when to buy it. Decisions are made from experience or simple rules of thumb. This causes three problems:

- **Poor purchasing decisions:** quantities are estimated by hand, leading to overstocking (wasted money and storage) or shortages (operational disruption).
- **Price uncertainty and fragmented information:** supplier prices, past purchases and stock levels are kept in separate files, so it is hard to tell whether a price is favorable or which supplier to buy from; unusual spending or sudden price increases are noticed only after the budget has been affected.
- **Generic tools ignore organizational behavior:** existing forecasting tools (e.g., SAP, or Oracle) are expensive, built for large commercial companies, and give the same style of recommendation to everyone. Each organization decides differently (for example, small stock buffers versus stocking up before holidays), so recommendations that ignore this habit are overridden and eventually ignored.

Why AI is needed: a person currently estimates future demand, judges whether a price change matters, and decides how much to buy and from which supplier. These are pattern-recognition tasks over historical data, which machine learning handles well. The system must also adapt to each organization's decision habits, which fixed formulas and reorder-point rules cannot do.

Who is affected: administrators, procurement officers and finance staff of educational institutions, which are the initial target of this project (for example, a university with many departments). The design can later be extended to businesses, NGOs and warehouses.

1.2 Purpose

The primary objectives of the software are:

- To provide a decision-support platform that helps organizations plan procurement and manage resources by learning from their own historical data and decisions.

- To forecast item-level demand, assess the risk of shortage or overstock, and display price trends so that administrators can see what they will need and how prices are moving.
- To recommend what to buy, how much, when and from which supplier, and to adapt those recommendations to each organization's past decisions (personalization).
- To evaluate whether organization-specific learning produces better recommendations than a generic model, by comparing three versions: generic, organization-history-based and personalized (central AI research goal).
- To detect unusual transactions early and explain why they were flagged (secondary objective).
- To provide a natural-language assistant, mainly to explain recommendations and answer questions grounded in the organization's own stored records (secondary objective).

1.3 Scope

The system will:

- Support organization setup (the initial release targets educational institutions) and role-based user accounts.
- Record and import (CSV) purchases, consumption, inventory, supplier and price data (transportation cost is kept only as an optional supplier attribute).
- Produce budget utilization and cost-per-department summaries for institutions.
- Forecast demand per item, assess inventory risk, analyze price trends and supplier prices, and recommend purchase quantity, timing and supplier.
- Learn from the gap between recommended and actual purchases (Organizational Memory / personalization) and compare the generic, organization-history and personalized models.
- Present a dashboard with charts and alerts. As secondary features, detect anomalous transactions and provide a natural-language assistant that explains recommendations and answers questions about the organization's data.

The system will not:

- Make or execute purchases or process payments (all AI output is a recommendation the user can accept or override).
- Track vehicles in real time or optimize GPS routes.

- Scrape market prices automatically (prices come from uploaded files or public datasets).
- Integrate with external ERP or accounting systems, provide a mobile app, or manage multiple branches.
- Guarantee future prices or replace the human decision maker.
- Provide profit/loss reporting, analyze transportation costs, or support business-type organizations in this release (future work).

System boundary: the system works only on data entered or uploaded by the organization, plus public datasets used for model development.

1.4 Core Focus and Research Question

Central question: To keep the project realistic and measurable, the system concentrates on one problem: can AI learn from an organization's historical purchasing behavior to recommend what, when and how much it should purchase? The initial system is designed for procurement and resource management in educational institutions, such as a university whose departments buy paper, toner, laboratory supplies and cleaning materials. Other organization types are future work.

Core workflow: Historical Data → Prediction → Recommendation → User Decision → Feedback → Organizational Memory → Improved Recommendation. Organizational Memory, the stored profile of how this organization actually decides, is the distinguishing feature of the system.

Illustrative example (paper): monthly paper consumption in six earlier months was 420, 450, 470, 620, 680 and 500 units. The forecast for next month is about 650 units and the current stock is 180 units, so the system recommends buying about 470 units from the supplier with the lower current price. The procurement officer can accept, modify or reject the recommendation; the decision is recorded and used to improve later recommendations.

Feature priorities:

- Core: users and roles; items, inventory, purchases, suppliers, consumption and price records; demand forecasting, inventory risk assessment, price trend analysis and procurement recommendation; Organizational Memory; and the evaluation of generic, organization-history and personalized models.
- Secondary (built only after the core works): anomaly detection and the natural-language assistant. The assistant is an interface on top of the data, used mainly to explain recommendations; it is not the main contribution.
- Deferred: profit/loss reporting for businesses and transportation cost analysis.

Known limitation: forecast quality depends on historical data. A new organization without enough history is served by the generic model first (cold start), and the system labels which model produced each result.

2. Design and Implementation Constraints

This section identifies the technical boundaries of the project.

2.1 Programming Language

Layer	Technology	Justification
Backend and ML	Python	Extensive libraries for data analysis, time-series forecasting and machine learning; one language for both the API and the AI components.
Frontend	JavaScript / TypeScript with React and a charting library (Recharts or Chart.js)	Component-based UI suited to interactive dashboards and charts.
Data access	SQL	Relational queries for transactions, inventory and reporting.

2.2 Database & Servers

- **PostgreSQL** stores relational data: organizations, users, items, transactions, inventory, prices, forecasts, recommendations, decisions and organization profiles (SQLite may be used during development).

- **FastAPI** (or Django) provides RESTful APIs that connect the React frontend, the AI services and the database.

- Data is entered through forms or imported from **CSV files**; no external ERP or accounting integration is used.

- The web application must run in modern web browsers; source control uses Git/GitHub.

2.3 AI Model & Services

AI Feature	Approach / Model	Why this choice
Demand forecasting, inventory risk and price trends	Time-series and regression models: Prophet or statsmodels, and gradient boosting (XGBoost or LightGBM)	Item consumption and prices are structured, seasonal time-series data; these models are fast, work on small datasets and give error ranges.
Procurement recommendation and Organizational Memory (personalization)	Forecast + stock + price trend rules, blended with a learned per-organization adjustment factor	Simple and explainable; the factor is learned from recommended-vs-actual gaps and falls back to the generic model when history is short (cold start).
Anomaly detection	Isolation Forest or statistical thresholds	Unsupervised: no labelled anomalies are needed by the organization.
Data-grounded assistant	LLM API (e.g., Claude API) with function calling over the database	Answers come from retrieved records, not the model's general knowledge, which reduces incorrect answers. Used mainly to explain recommendations (secondary feature).

The ML models are served as Python modules or endpoints behind the FastAPI backend. Training and experimentation use pandas, NumPy, scikit-learn and Jupyter Notebook.

3. User Classes and Characteristics

The system has the following user classes. All users can see only their own organization's data.

User Class	Characteristics	Permissions
System Administrator	Technical or senior staff who set up and maintain the organization's account.	Register the organization (type: educational institution in the initial release); create and manage user accounts and roles; configure the system and security settings.
Administrator / Manager	Main end users who review recommendations and make purchasing decisions. Not technical.	View dashboard, forecasts and price trends; review, accept or override recommendations; review flagged transactions; use the assistant; import data.
Procurement / Store Officer	Staff who handle day-to-day goods and stock records. Need fast, simple data entry.	Record and edit purchases, consumption/sales and inventory; import CSV data; view dashboard and stock alerts.
Finance / Accounts Staff	Staff who monitor expenses and budgets.	View financial summaries (budget utilization and cost per department), expense alerts and flagged transactions; use the assistant.

4. Requirement Elicitation

This section explains how the requirements were collected from real users and how they changed as a result.

4.1 Elicitation Techniques

Requirements were gathered in October 2026 from staff who handle purchasing, stock and budgets at the university's campus supershop and at Swapno, a supermarket chain, and were supported by a review of the purchase documents these organizations use. Both organizations buy goods from many suppliers and face the problem this system addresses: deciding what, how much and when to buy.

Technique	Participants / Source	What it covered
Client requirement questionnaire (paper form, sections A to J)	Finance / accounts staff of the university campus supershop, 4 to 10 years of experience. 1 completed form, dated 14 October 2026.	Current practice, data availability, supplier selection, trust and overrides, reports and roles, new organizations, assistants, and Must / Should / Could priorities for 11 features.
Face-to-face interviews	Staff at the university campus supershop and at Swapno.	Biggest problems, items causing most trouble, when demand rises, and how purchases are split between supplier types.
Document analysis	Six sample purchase documents photographed at the two organizations: four from the campus supershop (printed vendor bills, handwritten bills and an order memo) and two from Swapno (a purchase order and a handwritten cash memo).	Record formats, fields available, language of the records, and what CSV import and manual entry must handle.
Proposal and scope review	Project team	Check that the scope is realistic and the research question measurable (first four rows of Section 4.4).

4.2 Elicitation Evidence

- Completed Client Requirement Questionnaire, 7 pages, dated 14 October 2026 (Appendix A, Figures A.1 to A.7). The respondent is finance / accounts staff of the university campus supershop.
- Photographs of six purchase documents from the two organizations (Appendix A, Figures A.8 to A.13): printed vendor bills, handwritten bills and an order memo from the campus supershop, and a purchase order and a handwritten cash memo from Swapno.

4.3 Key Findings

The table summarises what users told us, grouped by topic, with the requirements each finding informed. Findings come from the questionnaire, the interviews and the sample documents.

Topic	Question area	Requirements informed	Finding (from users)
Current practice	Q1, Q3	FR6, FR9, FR11	Questionnaire: the quantity to buy is decided from the current stock level, buying follows a fixed schedule (monthly, quarterly or yearly), and ERP / accounting software is used. Interviews at the university supershop and at Swapno: the biggest problem for both is the risk of shortage or overstock, and missing products are reported most. Cosmetics are the most troublesome item group, and demand rises at the start of each month.
Data availability	Q2	FR3, FR4, FR5, NFR11	Purchase records and supplier details exist for 3 years or more and are mostly complete and consistent. They are kept on paper. Consumption/issue records and current stock levels were not marked as available. The respondent agreed to share a sample file. The sample bills mix printed vendor bills, handwritten bills and purchase orders, some in Bangla, and the printed bills have an empty unit-price column.

Topic	Question area	Requirements informed	Finding (from users)
Supplier selection	Q4	FR5, FR11, FR18	About 50% of purchases come from corporate suppliers and 50% from dealers. Suppliers are chosen by delivery time / reliability and quality of goods, and three or more suppliers are compared for an item. The respondent wants to see current prices side by side.
Trust and overrides	Q5	FR11, FR13, FR17, NFR6	Before trusting a recommendation, the respondent wants to see the price trend. A recommendation would be overridden when the budget is limited. Overrides should be learned only after the user confirms. The manager, procurement officer and department head should be able to accept or change a recommendation, and a reason should always be asked for an override.
Reports and roles	Q6	FR2, FR7, FR8, FR16, FR20, NFR5	Regular checks: current stock by item, and budget used and remaining. Alerts on the dashboard only. Each role should see only what it needs, and finance data should be restricted. Exports in CSV / Excel. Preferred devices: desktop / laptop and phone browsers. The chat assistant would be used "sometimes", for example to ask how much was spent on an item last month.
New organizations	Q7	FR14, NFR10	With little history, use a simple average of recent months. About 6 months of history are expected before recommendations are trusted.
Feature priorities	Q8	Priority column in Section 5	Must have record / import purchases and stock; shortage / overstock risk (written "V.V. I"); supplier price comparison; budget utilization. Should have demand forecast; learning from decisions. Could have purchase recommendation; dashboard and alerts. Not needed: price trends; unusual transaction alerts; chat assistant.

What would make users stop using the system: hard data entry, wrong forecasts and privacy concerns (questionnaire J2). These are addressed by simple manual entry and CSV import (FR3, FR4), data isolation (NFR3) and the evaluation of forecast accuracy (FR19).

Limits of this evidence: one completed questionnaire (finance / accounts staff, university supershop), interviews and six sample documents. Both organizations are retail buyers rather than university departments, but they face the same problem, and their examples (cosmetics, packaged food, eggs) will be used in the demonstration data.

4.4 Requirement Changes After Elicitation

The table lists requirements that were added, modified or removed. The first rows come from user input (questionnaire, interviews and sample documents). The last four rows come from the earlier review of the first proposal, before user input.

ID	Change	Before elicitation	After elicitation	Source / reason
FR18	Modified	Supplier Price Comparison: Medium; compared current and historical prices and showed the cheaper supplier.	High (Must have). Shows current prices of three or more suppliers side by side, with delivery time.	Questionnaire D2, D3, I; interview: about 50% of purchases from corporate suppliers and 50% from dealers.
FR7	Modified	Budget Utilization: Medium; only for institution-type organizations.	High (Must have). Budget used, remaining budget and cost per department, also shown on the dashboard.	Questionnaire F1 and I (Must have).
FR10	Modified	Price Trend Analysis: Medium, a separate feature.	Low. The price trend is shown inside each recommendation's reasoning instead of as a stand-alone feature.	Questionnaire I (Not needed) but E1 (price trend needed to trust a recommendation).
FR14	Modified	Cold start: use the generic model.	Use a simple average of recent months, labelled low confidence until about 6 months of history exist.	Questionnaire G1 and G2.

ID	Change	Before elicitation	After elicitation	Source / reason
FR13, UC-06	Modified	Learn from every recorded decision automatically.	Learn from a decision only after the user confirms it.	Questionnaire E3.
FR17, UC-05	Modified	The manager can override a recommendation; the reason is optional.	Manager, procurement officer or department head can accept, modify or reject; a reason is always required for an override.	Questionnaire E4 and E5.
FR2, NFR4	Modified	Roles: admin, manager, store officer.	Adds the department head; budget and expense data restricted to authorized roles.	Questionnaire E4 and F3.
FR3, FR4	Modified	Record purchases, consumption/sales and inventory; import CSV data.	Quick manual entry from paper bills with essential fields only; CSV import accepts spreadsheet and ERP exports.	Questionnaire C1, C3 and J2 (hard data entry); sample printed and handwritten bills.
FR5	Modified	Upload item price data.	Also upload supplier delivery time.	Questionnaire D1 (delivery time / reliability, quality).
FR9	Modified	Forecast next-period demand.	Monthly forecast that captures the rise in demand at the start of each month.	Interviews; questionnaire B2 and B6.
FR8	Modified	Dashboard of inventory, expenses and alerts.	Shows current stock by item, inventory risk, and budget used and remaining; alerts on the dashboard only.	Questionnaire F1 and F2.
FR20	New	None.	Export data and reports as CSV / Excel.	Questionnaire F4.

ID	Change	Before elicitation	After elicitation	Source / reason
NFR5	Modified	Usable on modern desktop browsers.	Also usable on phone browsers (responsive layout).	Questionnaire J1.
NFR11	New	None.	Item and supplier names accept Bangla and English text.	Sample documents: handwritten Bangla bills.
FR16	Modified	Assistant answers questions about the data.	Includes spending questions such as "How much did we spend on X last month?"; remains secondary (Low).	Questionnaire H1 and H2; the same form rated the chat assistant Not needed.
FR1, 1.3, 1.4	Modified	Initial scope: all organization types.	Educational institutions first; type field kept for later.	Proposal review (before user input): too many features and organization types for a capstone.
FR6, FR7	Removed / New	Profit/loss reporting and transportation cost analysis.	Both removed; inventory risk assessment added.	Proposal review: focus on what, when and how much to purchase.
FR15, FR16, NFR8, NFR9	Modified	Anomaly detection and the assistant as main features.	Secondary (Low); the assistant mainly explains recommendations.	Proposal review: supporting features, not the main contribution.
FR11, FR18, FR19, NFR7	New / Modified	No supplier choice and no way to measure personalization.	Recommendation includes the supplier; supplier price comparison and model comparison added.	Proposal review: the benefit of personalization must be measurable.

Kept as planned, although users ranked them lower: demand forecasting (FR9, "Should have"), procurement recommendation (FR11, "Could have") and learning from decisions (FR13, "Should have") stay High priority. The "Must have" shortage / overstock risk (FR6) depends on the forecast, the recommendation is how the risk becomes a purchase decision, and these three features are the project's central research question.

Kept secondary: anomaly detection (FR15) and the chat assistant (FR16) were rated "Not needed" in the questionnaire. They stay Low priority and will be built only after the core features work.

Confirmed without change: inventory risk assessment (FR6) was rated "Must have" and marked "V.V. I" by the respondent; recording and importing purchases and stock (FR3, FR4) was also "Must have".

5. Functional Requirements (FR)

ID	Requirement Name	Description	Users	Priority	Assignee
FR1	Organization Registration	The system must allow a user to register an organization. The initial release targets educational institutions; the type field is kept so other types can be added later.	System Admin	High	M1
FR2	Role-Based Accounts	The system must support role-based accounts (admin, manager, store officer) and restrict features by role.	System Admin	High	M1
FR3	Transaction & Inventory Entry	The system must allow users to record and edit purchases, consumption/sales and inventory.	Store Officer, Manager	High	M1
FR4	CSV Data Import	The system must allow users to import historical purchases, consumption and inventory data from CSV files, validating the format before saving.	Store Officer, Manager	High	M1
FR5	Price Data Upload	The system must allow users to upload item price data from CSV files.	Store Officer, Manager	Medium	M1
FR6	Inventory Risk Assessment	The system must estimate, for each item, the risk of shortage or overstock from the forecast, current stock and recent consumption, and show it as a risk level.	Manager	High	M2

ID	Requirement Name	Description	Users	Priority	Assignee
FR7	Budget Utilization	For institution-type organizations, the system must calculate budget utilization, remaining budget and cost per department.	Manager, Finance Staff	Medium	M1
FR8	Dashboard	The system must display a dashboard of inventory, expenses and alerts with charts.	All users	High	M3
FR9	Demand Forecasting	The system must forecast next-period demand for each item and display the forecast with an error range.	Manager	High	M2
FR10	Price Trend Analysis	The system must display the price trend direction and percentage change for each item that has price data.	Manager	Medium	M2
FR11	Procurement Recommendation	The system must recommend a purchase quantity, timing and supplier for each item based on forecast, current stock, price trend and supplier prices.	Manager	High	M2
FR12	Decision Recording	The system must record the user's final decision (accepted quantity or override) against each recommendation.	Manager	High	M1, M2

ID	Requirement Name	Description	Users	Priority	Assignee
FR13	Personalization (Organizational Memory)	The system must update an organization-specific adjustment from recorded decisions and use it in later recommendations.	Manager (indirect)	High	M2
FR14	Cold-Start Fallback	The system must use the generic model when an organization has insufficient history, and indicate which model was used.	Manager	High	M2
FR15	Anomaly Detection (secondary)	The system must flag transactions that deviate significantly from historical patterns and show the reason for each flag.	Manager, Finance Staff	Low	M3
FR16	Data-Grounded Assistant (secondary)	The system must answer natural-language questions about the organization's data, including why a recommendation was made, using only that organization's records and say so when data is insufficient.	Manager, Finance Staff	Low	M3
FR17	Recommendation Override	The system must allow the user to override any AI recommendation.	Manager	High	M3, M1
FR18	Supplier Price Comparison	The system must compare the current and historical prices of an item across its suppliers and indicate which supplier is currently cheaper.	Manager	Medium	M2

ID	Requirement Name	Description	Users	Priority	Assignee
FR19	Model Comparison and Evaluation	The system must evaluate and report the error (e.g., MAE) of the generic, organization-history and personalized models on held-out data so that the effect of personalization can be measured.	System Admin, Manager	High	M2

6. Non-Functional Requirements

Category	ID	Requirement
Performance	NFR1	Dashboard pages must be loaded within 5 seconds for datasets of up to 10,000 records.
Performance	NFR2	Assistant responses must be returned within 30 seconds under normal conditions.
Security & Privacy	NFR3	Each organization's data must be restricted to its own authorized users; no query or AI call may return another organization's records.
Security & Privacy	NFR4	Access to features must be controlled by user role.
Usability	NFR5	The system must be usable on modern web browsers (current versions of Chrome, Firefox, Edge).
Usability	NFR6	Each AI recommendation must show the reasoning behind it (forecast, stock, price trend) so users can judge it and override it.
AI Quality	NFR7	The system must compare the generic, organization-history and personalized models on held-out simulated test data using MAE and report whether personalization improves accuracy. The target is a lower MAE for the personalized model on simulated organizations with known behavior; if the target is not met, the result is reported as a finding.
AI Quality	NFR8	(Secondary feature) The anomaly detector should flag injected anomalies in the test data with at least 70% recall.
AI Quality	NFR9	(Secondary feature) The assistant must answer only from retrieved records and state when data is insufficient. If the LLM service is unavailable, the system must show a clear message, and all other features must keep working.
AI Quality	NFR10	If forecasting cannot run (insufficient history or model failure), the system must fall back to the generic model or a simple moving-average estimate and label the result accordingly.

7. System Design Diagrams

7.1 Use Case Diagram

The diagram shows how the four user classes and the external LLM service interact with the system. "Personalize Recommendations" runs automatically when a decision is recorded.

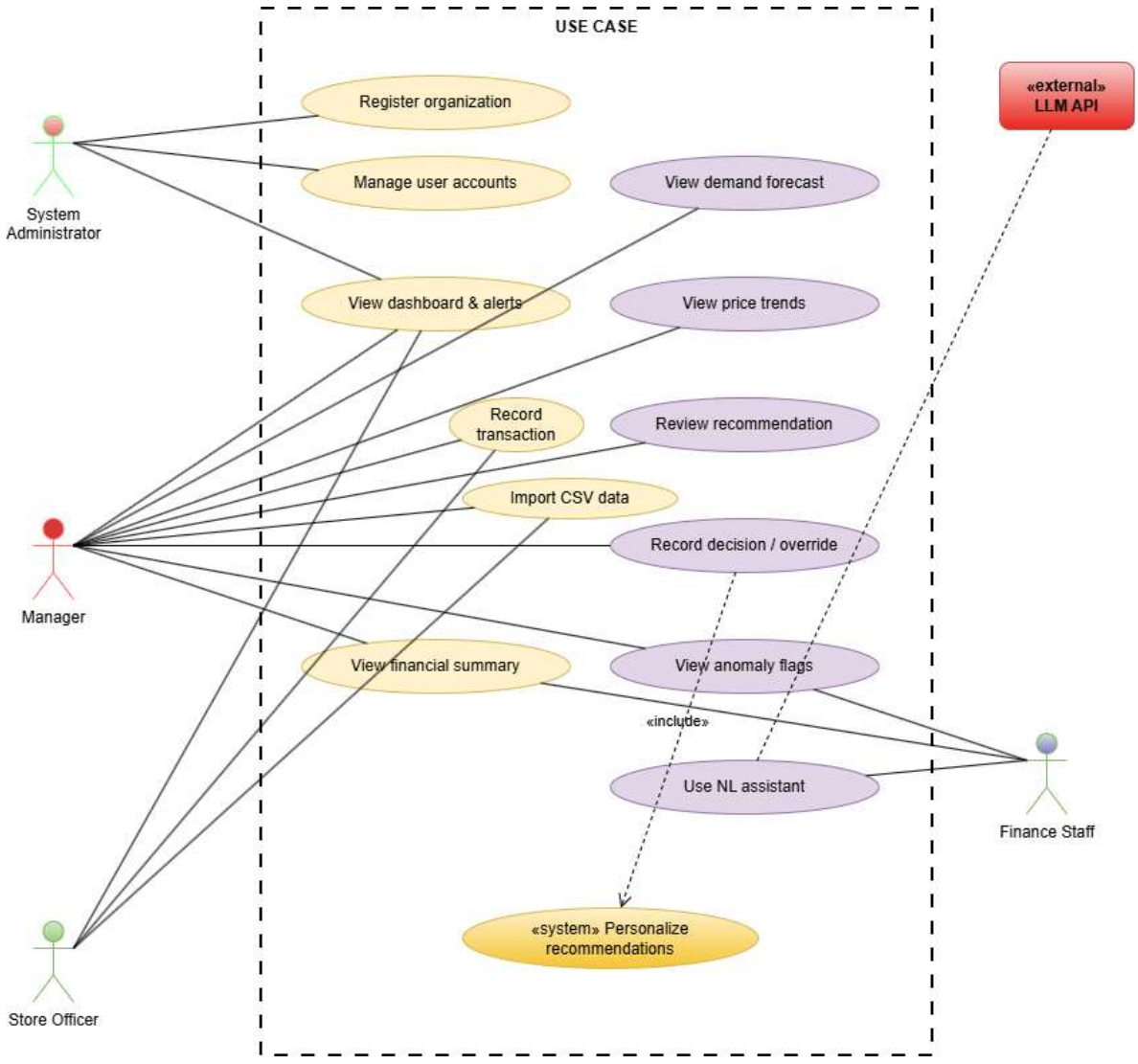


Figure 7.1: Use case diagram

7.2 Use Case Descriptions

Main Success Flows for the High-priority features.

Use Case - 01: Register Organization and Manage Accounts

Actor(s)	System Administrator
Related requirements	FR1, FR2
Preconditions	None; the administrator can open the registration page.
Main Success Flow	1. Administrator opens the registration page and enters the organization's details. 2. Administrator confirms the organization type (educational institution in the initial release). 3. System validates the input and creates the organization with its own isolated data space. 4. Administrator creates user accounts and assigns each a role (admin, manager, store officer). 5. System saves the accounts and confirms; users can now log in.
Alternative / Exception Flows	3a. Required field missing or organization already registered: system shows an error and nothing is saved. 4a. Invalid or duplicate email: system rejects the account and asks for a correction.
Postconditions	Organization and role-based accounts exist; the organization type is stored for future extension.

Use Case - 02: Record or Edit Purchases, Consumption and Inventory

Actor(s)	Procurement / Store Officer, Manager
Related requirements	FR3
Preconditions	User is logged in with permission to enter data.

Main Success Flow	1. User opens the data entry page and chooses the record type (purchase, consumption/sales or inventory). 2. User enters item, quantity, price/cost, supplier, department and date. 3. System validates the entries. 4. System saves the record against the organization. 5. System updates stock levels and confirms to the user.
Alternative / Exception Flows	3a. Invalid values (negative quantity, missing item): system highlights the fields and does not save. E1. User edits an existing record: system saves the change and recalculates stock.
Postconditions	Records are stored and available for dashboard, forecasting and anomaly checks.

Use Case - 03: Import Historical Data from CSV

Actor(s)	Procurement / Store Officer, Manager
Related requirements	FR4, FR5
Preconditions	User is logged in; a CSV file with the expected columns is available.
Main Success Flow	1. User opens the import page and selects the data type (purchases, consumption/sales, inventory or prices). 2. User uploads the CSV file. 3. System validates the columns and value formats. 4. System shows a preview with the number of rows found. 5. User confirms the import. 6. System saves the records and shows an import summary.
Alternative / Exception Flows	3a. Format errors: system lists the faulty rows and saves nothing. 5a. User cancels: no data is saved.
Postconditions	Historical data is stored and used by the forecasting and anomaly modules.

Use Case - 04: View Demand Forecast

Actor(s)	Manager
Related requirements	FR9, FR14
Preconditions	User is logged in; the item has recorded consumption history.
Main Success Flow	1. Manager selects an item and a forecast period. 2. System retrieves the item's consumption history. 3. System checks whether there is enough history and selects the organization-trained or the generic model. 4. System computes the predicted demand with an error range. 5. System displays the forecast chart, the error range and which model was used.
Alternative / Exception Flows	3a. Insufficient history: system uses the generic model and labels the result "generic". 2a. No data at all: system asks the user to enter or import data first.
Postconditions	Forecast is stored and used by the recommendation engine.

Use case - 05: Review Recommendation and Record Decision

Actor(s)	Manager
Related requirements	FR11, FR12, FR17
Preconditions	A forecast and current stock exist for the item.
Main Success Flow	1. Manager opens the recommendations list. 2. System shows, per item, the recommended quantity, timing and supplier with the forecast, stock level, inventory risk and price trend behind it. 3. Manager selects an item and reviews the reasoning. 4. Manager accepts the recommendation. 5. System records the decision (final quantity = recommended quantity). 6. System triggers personalization (UC-06).

Alternative / Exception Flows	4a. Override: Manager enters a different quantity (optional reason); system records it as the final decision. 4b. Manager postpones: no decision is recorded and the recommendation stays open.
Postconditions	Final decision is stored against the recommendation. No purchase is executed by the system.

Use Case - 06: Personalize Recommendations

Actor(s)	System (triggered by a recorded decision)
Related requirements	FR13, FR14
Preconditions	At least one recorded decision exists for the item.
Main Success Flow	1. System reads the new decision and the recommendation it answers. 2. System computes the ratio of the final quantity to the recommended quantity. 3. System blends this ratio into the item's stored adjustment factor. 4. System saves the updated factor and decision count in the organization profile. 5. The next recommendation for this item applies to the updated factor.
Alternative / Exception Flows	3a. Fewer than 5 recorded decisions: the factor is stored but not yet applied; the generic recommendation is used.
Postconditions	Organization profile is updated; recommendations gradually match the organization's habits.

Use Case - 07: View Dashboard and Alerts

Actor(s)	All users
Related requirements	FR8
Preconditions	User is logged in.

Main Success Flow	1. User opens the dashboard. 2. System loads inventory levels, expenses and open alerts for the user's organization. 3. System displays charts and an alert list (low stock, shortage or overstock risk, overspending, flagged transactions). 4. User selects an alert to view its details.
Alternative / Exception Flows	2a. No data yet: system shows an empty-state message with a link to the import page.
Postconditions	Users are informed of the organization's current state.

7.3 Activity Diagram

The diagram maps the procedural flow of the core feature, "Procurement Recommendation and Decision", from the forecast to the personalization feedback loop and anomaly check. Green steps are AI components.

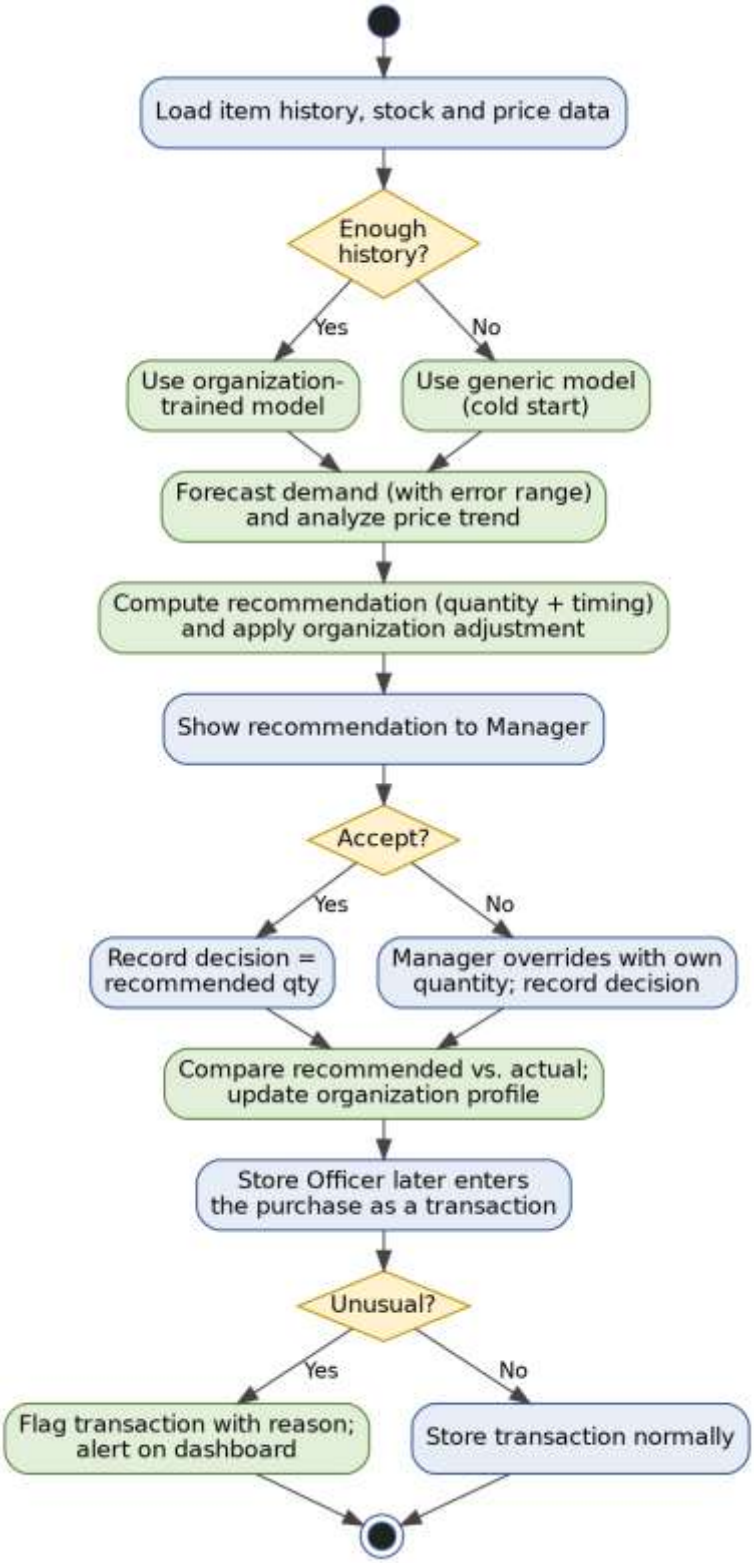


Figure 7.3: Activity diagram, procurement recommendation and decision

7.4 Entity Relationship Diagram (ERD)

PK = primary key, FK = foreign key. Every entity is scoped to an organization, which supports the data isolation requirement (NFR3).

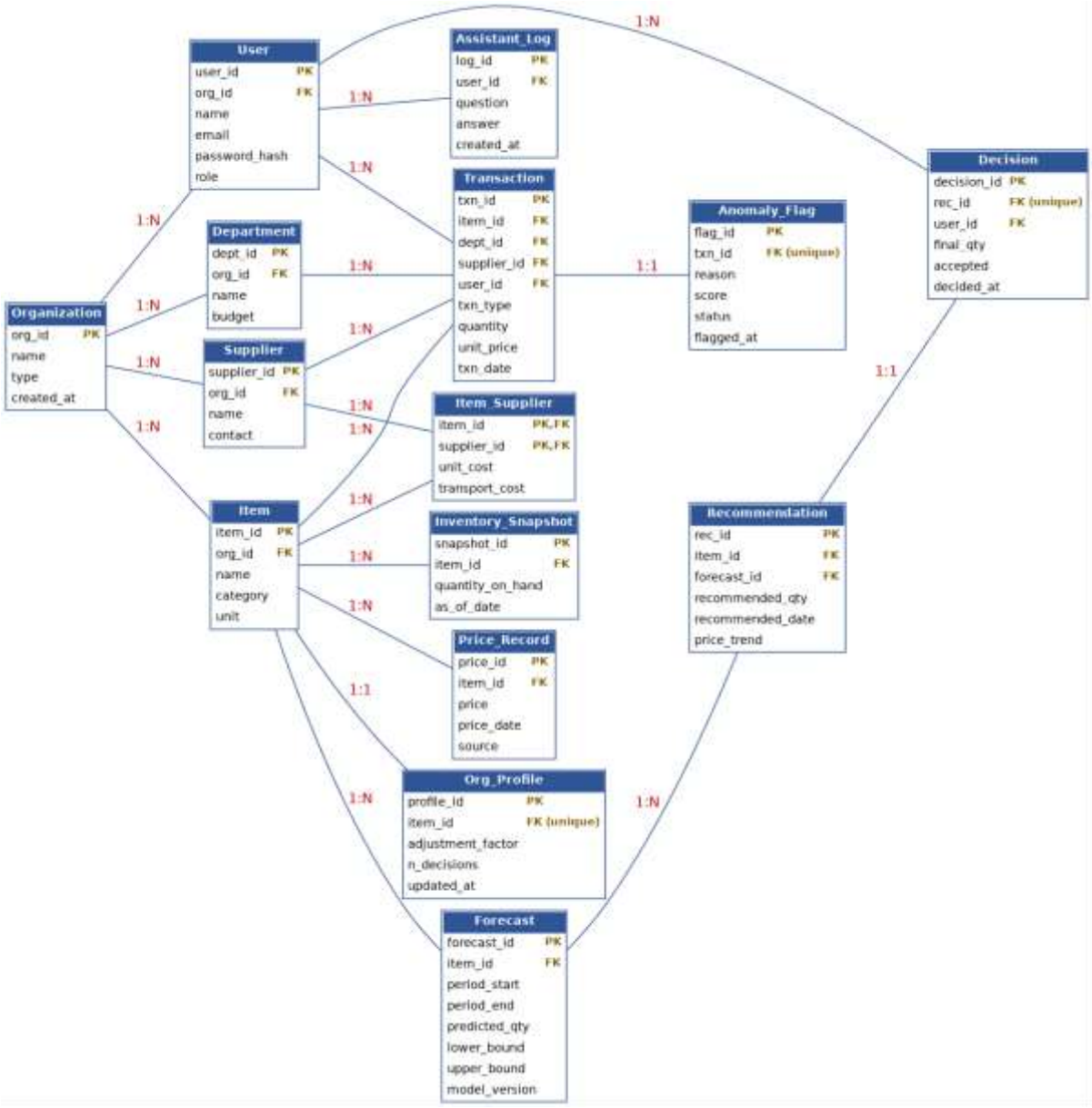


Figure 7.4: Entity relationship diagram

Relationships:

Entity A	Entity B	Type	Description
Organization	User	1:N	One organization has many users; each user belongs to one organization.
Organization	Department	1:N	One organization has many departments (used for institutions).
Organization	Item / Supplier	1:N	One organization owns many items and many suppliers.
Item	Supplier	M: N	An item can come from many suppliers and a supplier can supply many items; resolved by Item_Supplier (holds unit price and optional transport cost).
Item	Transaction	1: N	One item appears in many purchase/consumption transactions.
Department / Supplier / User	Transaction	1:N	A transaction may be linked to one department, supplier and recording user (department and supplier are optional).
Item	Inventory_Snapshot	1:N	One item has many stock snapshots over time.
Item	Price_Record	1:N	One item has many price records over time.
Item	Forecast	1:N	One item has many forecasts (one per period).
Forecast	Recommendation	1:N	One forecast can support recommendations (for different stock levels or dates).
Recommendation	Decision	1:1	Each recommendation has at most one final decision.

Entity A	Entity B	Type	Description
User	Decision	1:N	A user can make many decisions.
Item	Org_Profile	1:1	Each item has one personalization profile (adjustment factor) per organization.
Transaction	Anomaly_Flag	1:1	A transaction has at most one anomaly flag.
User	Assistant_Log	1:N	A user can ask many assistant questions; each is logged.

7.5 AI Feature Design

The AI layer is the core of the product. It turns stored operational data into forecasts, inventory risk, price trends and procurement recommendations, and learns from user decisions (Organizational Memory). Anomaly detection and the assistant are secondary components.

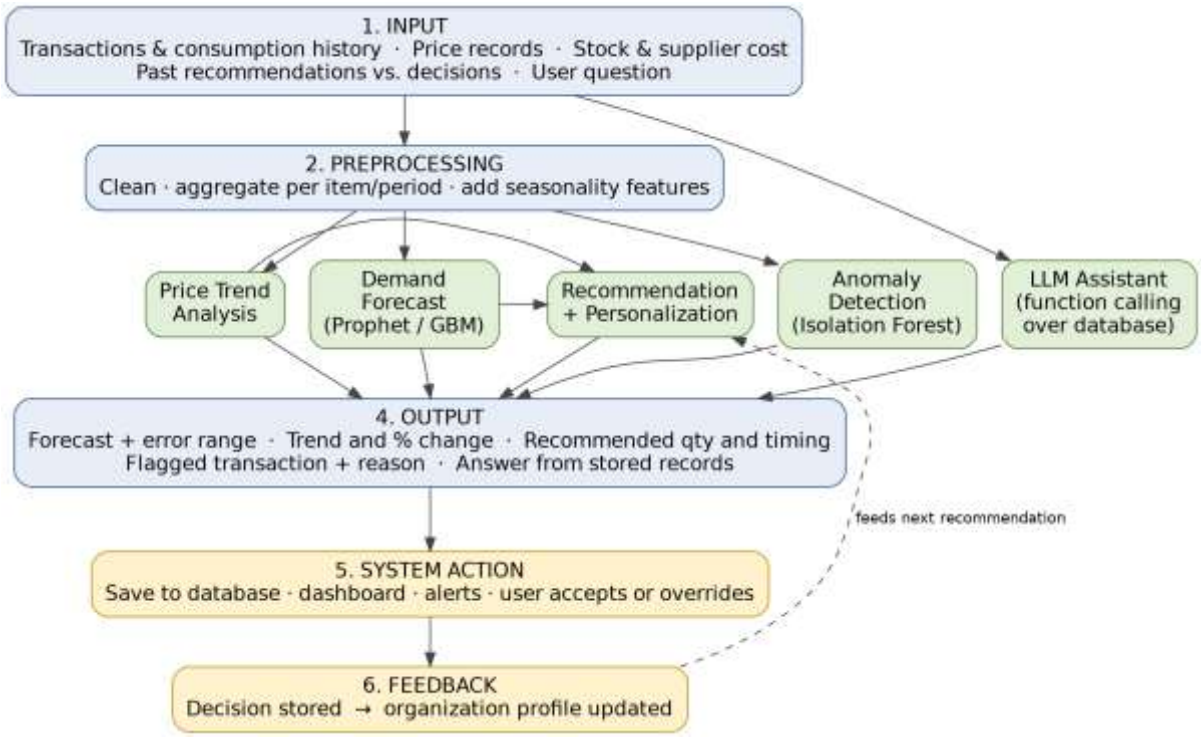


Figure 7.5: AI pipeline

Data

Source	Public retail/demand datasets (e.g., from Kaggle) for the generic model, plus simulated educational-institution organizations (departments buying paper, toner, laboratory and cleaning supplies) with defined purchasing behavior (e.g., one organization always buys 25% below the recommendation). In use, each organization's own entered or imported data.
Size	3 to 5 simulated organizations, about 50 to 200 items each, 36 months' worth of data for history. Updates when datasets are chosen.
Labelling	Forecasting: actual consumption in the next period is the label (no manual labelling). Personalization: the label is the user's actual final decision. Anomaly detection: unsupervised; known anomalies are injected into simulated data only to measure recall.
Preparation	Remove duplicates and invalid rows, aggregate per item and period, add calendar and seasonality features, split by time (the latest months are held out as test data).

Model

Feature	Approach	Why it fits	Baseline
Demand forecasting	Prophet / statsmodels or gradient boosting	Seasonal time-series; gives error range	Moving average of recent periods
Price trend	Trend estimation on price history (regression / time-series)	Direction and change are easy to explain	Simple price difference
Inventory risk	Expected demand compared with current stock and a safety buffer; output is a low / medium / high risk level	Simple and explainable to procurement staff	Fixed reorder-point rule
Procurement recommendation + Organizational Memory	Forecast, stock and price trend combined, then scaled by a learned organization factor	Adapts to each organization's habits; explainable	Generic model; model trained in organization history only

Feature	Approach	Why it fits	Baseline
Anomaly detection (secondary)	Isolation Forest or statistical thresholds	No labels required	Fixed z-score threshold
Assistant (secondary)	LLM API with function calling over the database	Answers come from real records	LLM without database access

Pipeline

Input	Transaction and consumption history, price records, current stock and supplier cost, past recommendations vs. actual decisions, user questions.
Preprocessing	Cleaning, aggregation per item/period, calendar and seasonality features.
Model	Forecast, inventory risk, price trend and recommendation with Organizational Memory; secondary: anomaly detection and LLM assistant (Section 2.3).
Output	Forecast with error range; inventory risk level; trend direction and % change; recommended quantity, timing and supplier; flagged transaction with reason; answer from stored records.
System action	Results saved to the database and shown on the dashboard; alerts raised for anomalies; user accepts or overrides; the decision is stored and the organization profile updated.

Integration

- A background job runs the forecast, inventory-risk, price-trend and recommendation modules after new data is imported and on a regular schedule and saves results to the Forecast and Recommendation tables.
- (Secondary feature) When a transaction is saved, the backend calls the anomaly detector and stores any result in Anomaly_Flag.
- (Secondary feature) For assistant questions, the backend sends the question to the LLM API with a fixed set of database functions limited to the user's organization; the model answers from the returned records, and the exchange is stored in Assistant_Log.
- When the user records a decision, the personalization module updates the Org_Profile table.

Evaluation & Fallback

Component	Metrics and test data	If the AI is wrong or unavailable
Forecasting	MAE and MAPE on held-out months, compared with the moving-average baseline	Use generic model or moving average; label the result.
Recommendation + Organizational Memory	Compare three versions (generic, organization history, personalized); measure how closely recommendations match the organization's actual purchases and whether the known simulated behavior (e.g., buys 25% below) is recovered	User can always override; generic recommendation is used until enough decisions exist.
Anomaly detection (secondary)	Precision and recall on injected anomalies in simulated data	Users can ignore a flag; thresholds can be adjusted. If the detector fails, transactions are saved without a flag.
Assistant (secondary)	Accuracy on a fixed set of test questions with known answers from the database (about 30 questions)	States when data is insufficient; if the LLM API is down, shows a message and the rest of the system continues to work.

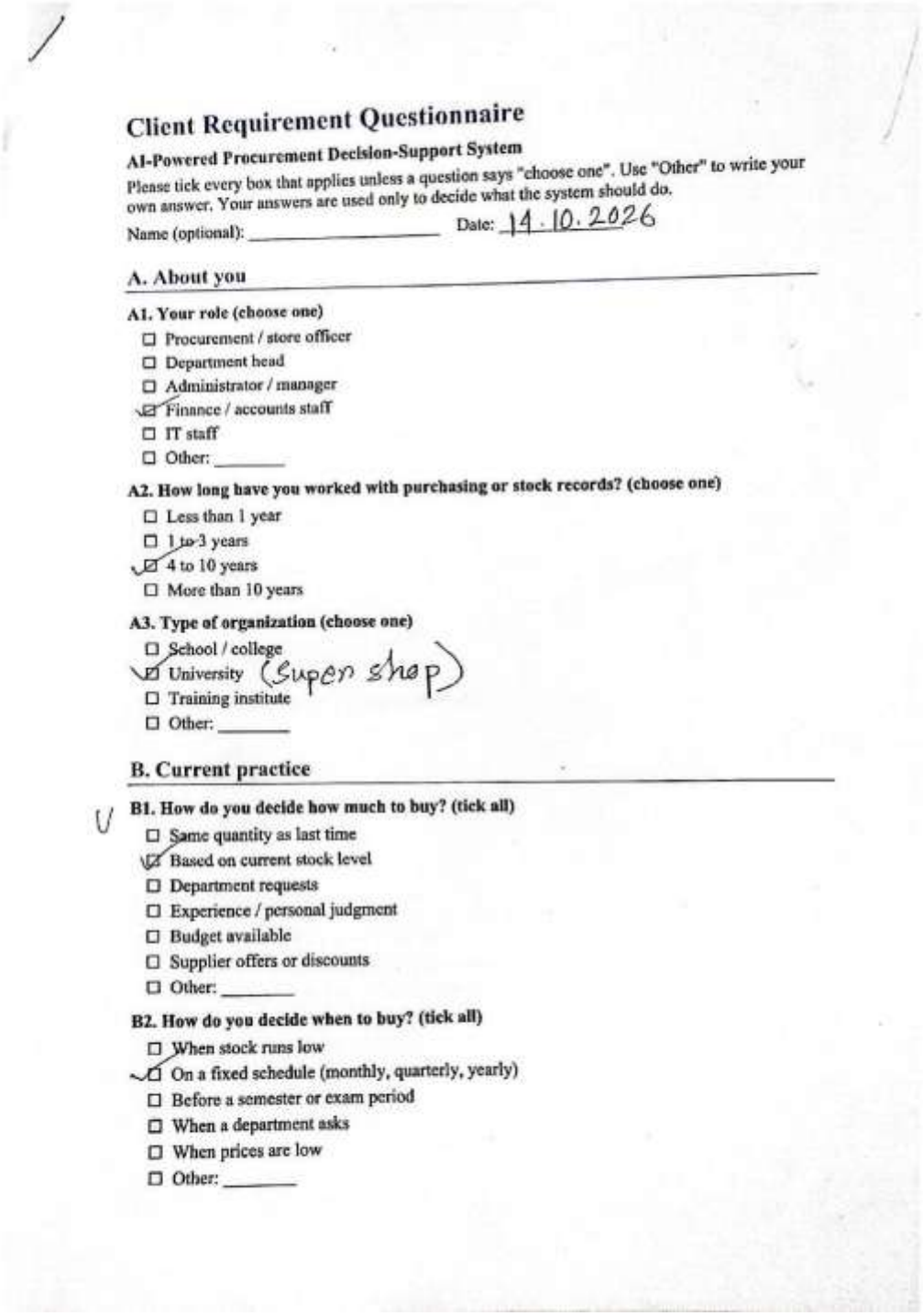
8. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
MD Asif Raihan	241-35-340	Backend API development, database schema, authentication, budget utilization module, system integration.
MD Abu Musahid Rony	241-35-273	AI model, data pipeline and evaluation: demand forecasting, inventory risk, price analysis, recommendation, Organizational Memory (personalization), evaluation of generic vs organization vs personalized models, simulated data.
MD Mostahid Islam Riad	241-35-398	Frontend UI/UX and dashboard, secondary features (anomaly detection, LLM assistant), testing and documentation.

All members take part in requirements, testing and the final report.

Appendix A: Elicitation Evidence

Evidence collected from the university campus supershop and from Swapno in October 2026. Figures A.1 to A.7 are the completed Client Requirement Questionnaire. Figures A.8 to A.13 are photographs of the purchase documents the organizations use.



Client Requirement Questionnaire

AI-Powered Procurement Decision-Support System

Please tick every box that applies unless a question says "choose one". Use "Other" to write your own answer. Your answers are used only to decide what the system should do.

Name (optional): _____ Date: 14.10.2026

A. About you

A1. Your role (choose one)

- ☐ Procurement / store officer
- ☐ Department head
- ☐ Administrator / manager
- ☑ Finance / accounts staff
- ☐ IT staff
- ☐ Other: _____

A2. How long have you worked with purchasing or stock records? (choose one)

- ☐ Less than 1 year
- ☐ 1 to 3 years
- ☑ 4 to 10 years
- ☐ More than 10 years

A3. Type of organization (choose one)

- ☐ School / college
- ☑ University (Super shop)
- ☐ Training institute
- ☐ Other: _____

B. Current practice

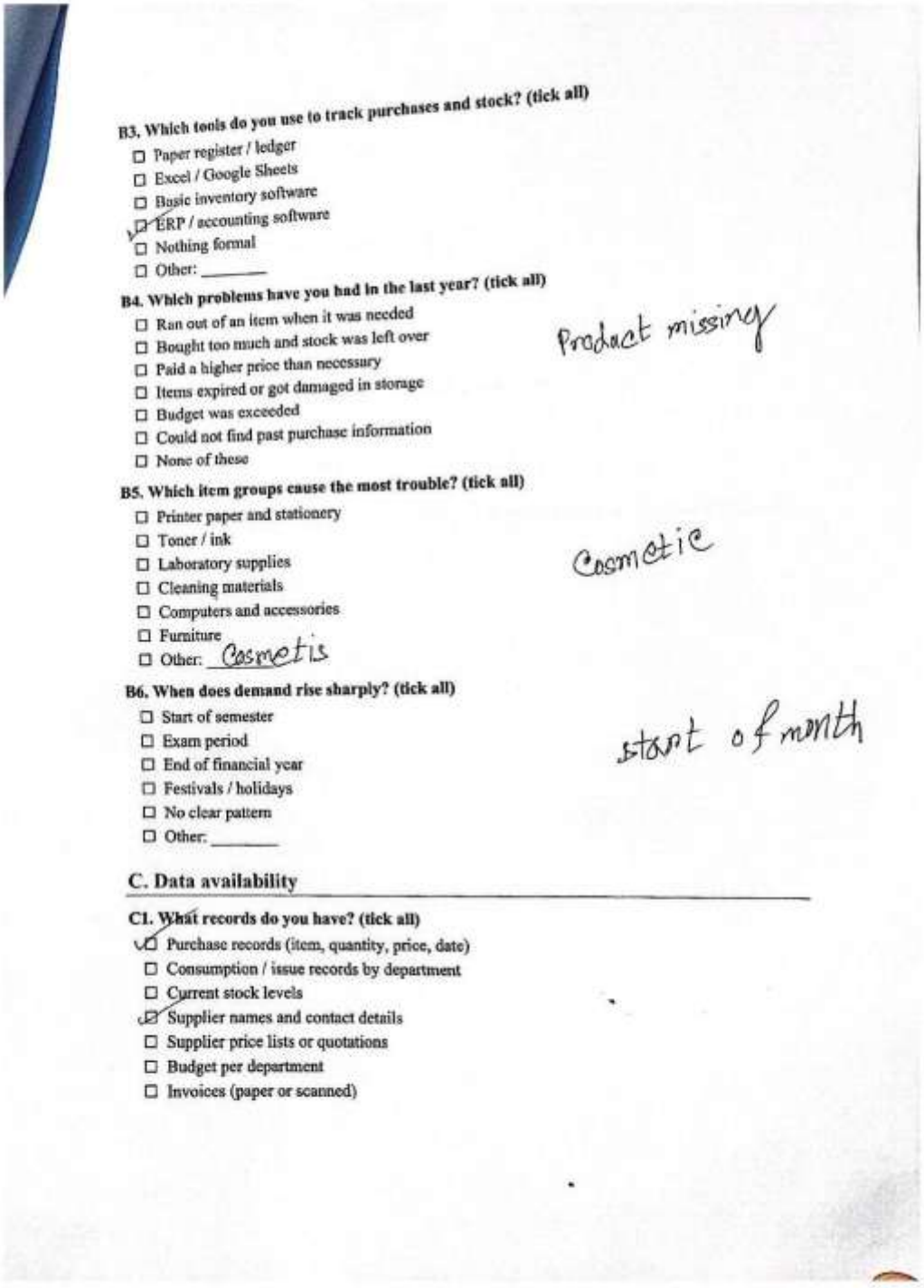
B1. How do you decide how much to buy? (tick all)

- ☐ Same quantity as last time
- ☑ Based on current stock level
- ☐ Department requests
- ☐ Experience / personal judgment
- ☐ Budget available
- ☐ Supplier offers or discounts
- ☐ Other: _____

B2. How do you decide when to buy? (tick all)

- ☐ When stock runs low
- ☑ On a fixed schedule (monthly, quarterly, yearly)
- ☐ Before a semester or exam period
- ☐ When a department asks
- ☐ When prices are low
- ☐ Other: _____

Figure A.1: Client Requirement Questionnaire, page 1 of 7 (completed 14 October 2026)



B3. Which tools do you use to track purchases and stock? (tick all)

- ☐ Paper register / ledger
- ☐ Excel / Google Sheets
- ☐ Basic inventory software
- ☑ ERP / accounting software
- ☐ Nothing formal
- ☐ Other: _____

B4. Which problems have you had in the last year? (tick all)

- ☐ Ran out of an item when it was needed
- ☐ Bought too much and stock was left over
- ☐ Paid a higher price than necessary
- ☐ Items expired or got damaged in storage
- ☐ Budget was exceeded
- ☐ Could not find past purchase information
- ☐ None of these

Product missing

B5. Which item groups cause the most trouble? (tick all)

- ☐ Printer paper and stationery
- ☐ Toner / ink
- ☐ Laboratory supplies
- ☐ Cleaning materials
- ☐ Computers and accessories
- ☐ Furniture
- ☐ Other: Cosmetis

Cosmetic

B6. When does demand rise sharply? (tick all)

- ☐ Start of semester
- ☐ Exam period
- ☐ End of financial year
- ☐ Festivals / holidays
- ☐ No clear pattern
- ☐ Other: _____

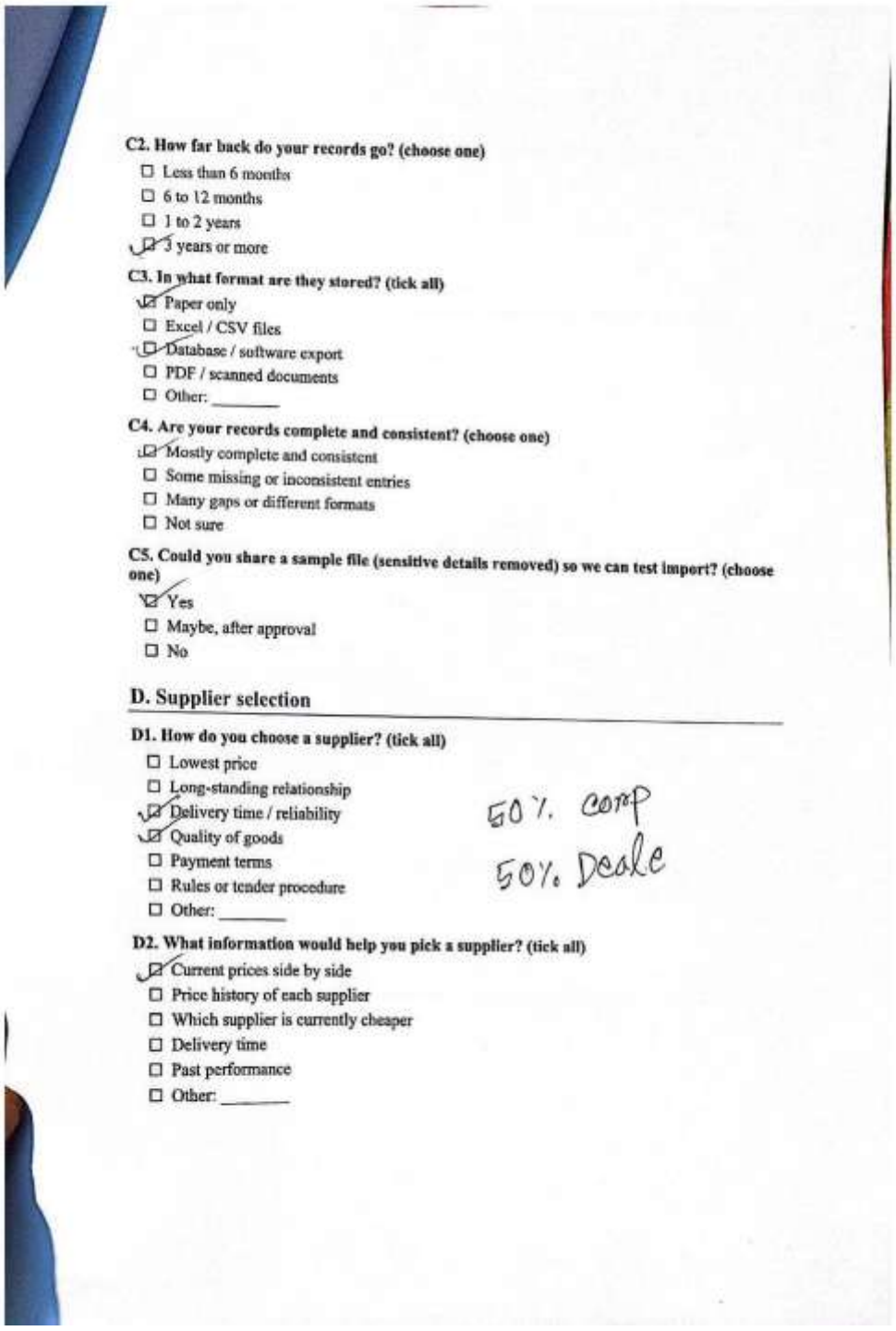
start of month

C. Data availability

C1. What records do you have? (tick all)

- ☑ Purchase records (item, quantity, price, date)
- ☐ Consumption / issue records by department
- ☐ Current stock levels
- ☑ Supplier names and contact details
- ☐ Supplier price lists or quotations
- ☐ Budget per department
- ☐ Invoices (paper or scanned)

Figure A.2: Client Requirement Questionnaire, page 2 of 7 (completed 14 October 2026)



C2. How far back do your records go? (choose one)

- ☐ Less than 6 months
- ☐ 6 to 12 months
- ☐ 1 to 2 years
- ☑ 3 years or more

C3. In what format are they stored? (tick all)

- ☑ Paper only
- ☐ Excel / CSV files
- ☐ Database / software export
- ☐ PDF / scanned documents
- ☐ Other: _____

C4. Are your records complete and consistent? (choose one)

- ☑ Mostly complete and consistent
- ☐ Some missing or inconsistent entries
- ☐ Many gaps or different formats
- ☐ Not sure

C5. Could you share a sample file (sensitive details removed) so we can test import? (choose one)

- ☑ Yes
- ☐ Maybe, after approval
- ☐ No

D. Supplier selection

D1. How do you choose a supplier? (tick all)

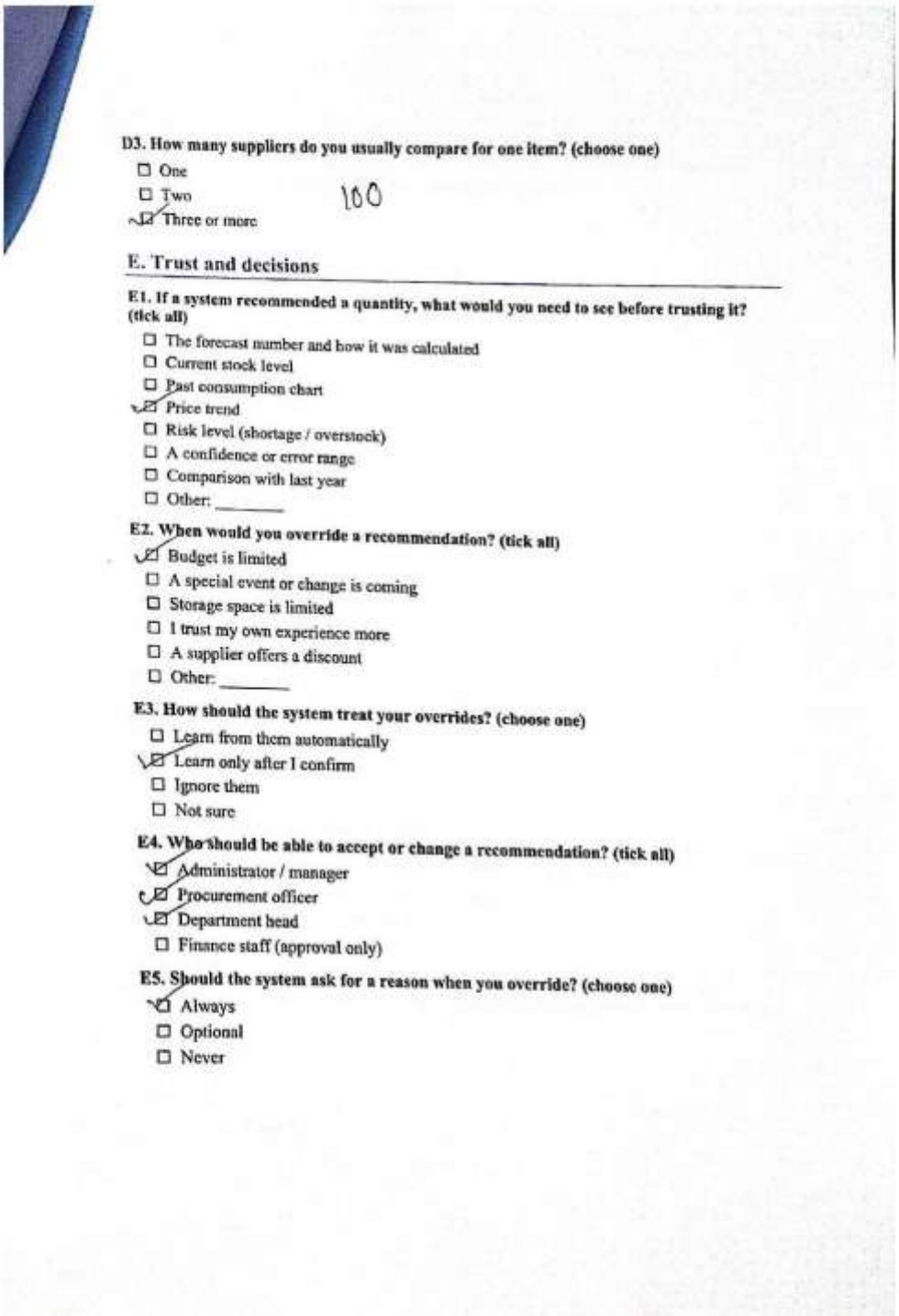
- ☐ Lowest price
- ☐ Long-standing relationship
- ☑ Delivery time / reliability
- ☑ Quality of goods
- ☐ Payment terms
- ☐ Rules or tender procedure
- ☐ Other: _____

50% comp
50% Deale

D2. What information would help you pick a supplier? (tick all)

- ☑ Current prices side by side
- ☐ Price history of each supplier
- ☐ Which supplier is currently cheaper
- ☐ Delivery time
- ☐ Past performance
- ☐ Other: _____

Figure A.3: Client Requirement Questionnaire, page 3 of 7 (completed 14 October 2026)



D3. How many suppliers do you usually compare for one item? (choose one)

- ☐ One
- ☐ Two
- ☑ Three or more

100

E. Trust and decisions

E1. If a system recommended a quantity, what would you need to see before trusting it? (tick all)

- ☐ The forecast number and how it was calculated
- ☐ Current stock level
- ☐ Past consumption chart
- ☑ Price trend
- ☐ Risk level (shortage / overstock)
- ☐ A confidence or error range
- ☐ Comparison with last year
- ☐ Other: _____

E2. When would you override a recommendation? (tick all)

- ☑ Budget is limited
- ☐ A special event or change is coming
- ☐ Storage space is limited
- ☐ I trust my own experience more
- ☐ A supplier offers a discount
- ☐ Other: _____

E3. How should the system treat your overrides? (choose one)

- ☐ Learn from them automatically
- ☑ Learn only after I confirm
- ☐ Ignore them
- ☐ Not sure

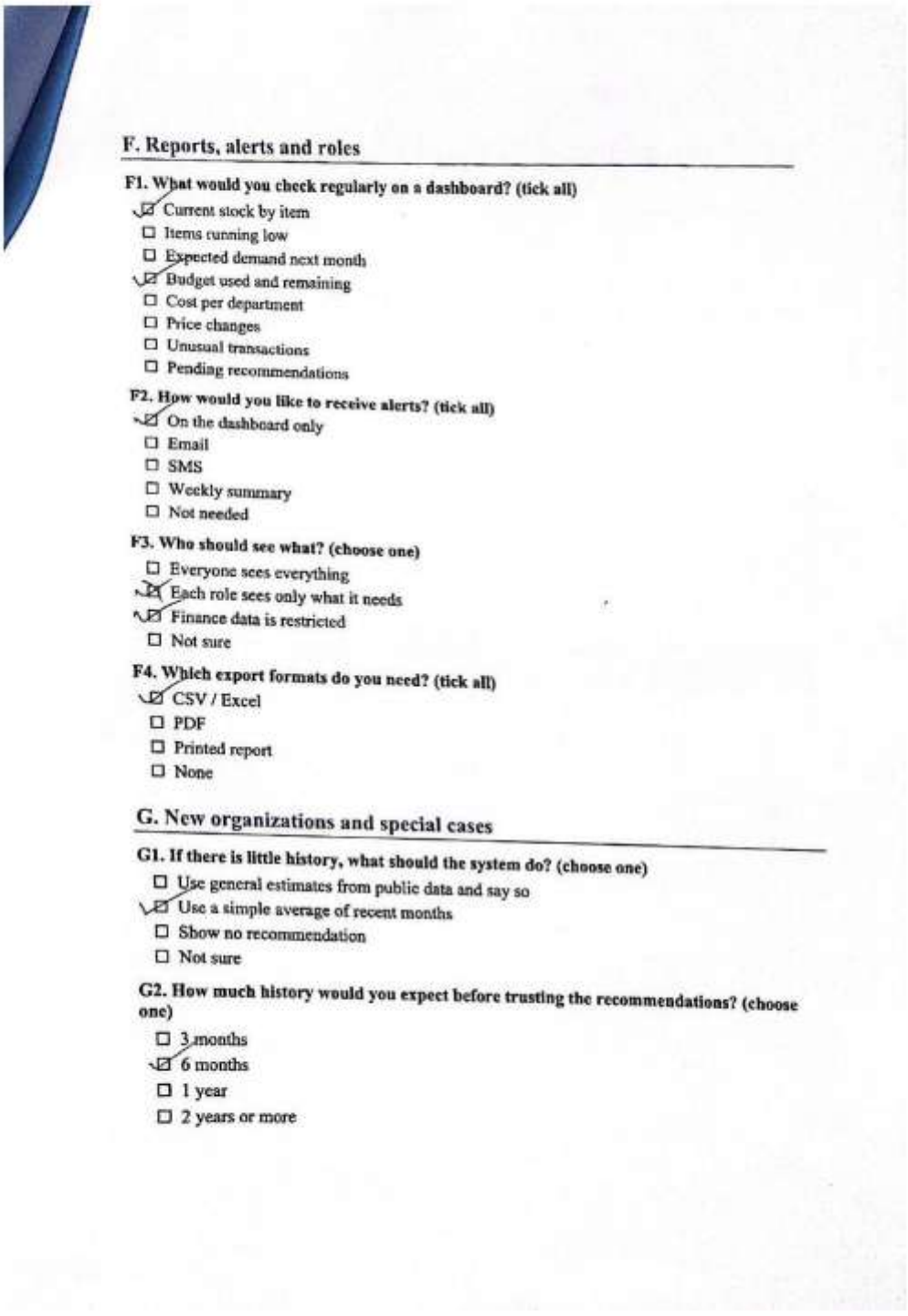
E4. Who should be able to accept or change a recommendation? (tick all)

- ☑ Administrator / manager
- ☑ Procurement officer
- ☑ Department head
- ☐ Finance staff (approval only)

E5. Should the system ask for a reason when you override? (choose one)

- ☑ Always
- ☐ Optional
- ☐ Never

Figure A.4: Client Requirement Questionnaire, page 4 of 7 (completed 14 October 2026)



F. Reports, alerts and roles

F1. What would you check regularly on a dashboard? (tick all)

- [x] Current stock by item
- [ ] Items running low
- [ ] Expected demand next month
- [x] Budget used and remaining
- [ ] Cost per department
- [ ] Price changes
- [ ] Unusual transactions
- [ ] Pending recommendations

F2. How would you like to receive alerts? (tick all)

- [x] On the dashboard only
- [ ] Email
- [ ] SMS
- [ ] Weekly summary
- [ ] Not needed

F3. Who should see what? (choose one)

- [ ] Everyone sees everything
- [x] Each role sees only what it needs
- [x] Finance data is restricted
- [ ] Not sure

F4. Which export formats do you need? (tick all)

- [x] CSV / Excel
- [ ] PDF
- [ ] Printed report
- [ ] None

G. New organizations and special cases

G1. If there is little history, what should the system do? (choose one)

- [ ] Use general estimates from public data and say so
- [x] Use a simple average of recent months
- [ ] Show no recommendation
- [ ] Not sure

G2. How much history would you expect before trusting the recommendations? (choose one)

- [ ] 3 months
- [x] 6 months
- [ ] 1 year
- [ ] 2 years or more

Figure A.5: Client Requirement Questionnaire, page 5 of 7 (completed 14 October 2026)

G3. How should the system handle one-off events (new lab, new course, big event)? (tick all)

- ☐ Let me mark them as special
- ☐ Ignore them in forecasts
- ☐ Include them as normal
- ☐ Not sure

H. Natural-language assistant (secondary)

H1. Would you use a chat assistant? (choose one)

- ☐ Yes, often
- ☑ Sometimes
- ☐ No

H2. What would you ask it? (tick all)

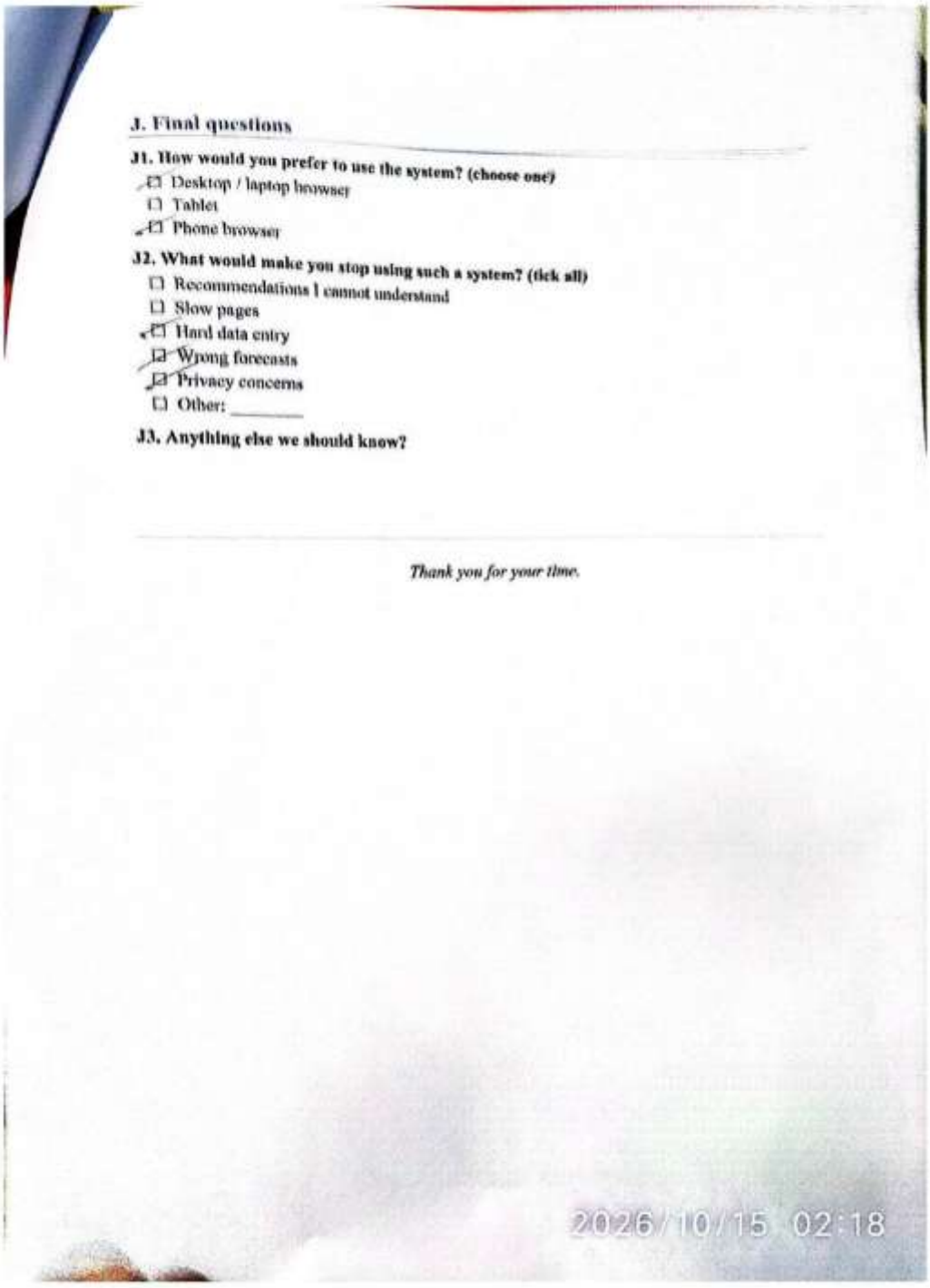
- ☐ Why was this quantity recommended?
- ☑ How much did we spend on X last month?
- ☐ Which items are running low?
- ☐ Which supplier was cheapest?
- ☐ Other: _____

I. Feature priority

For each feature, tick one box.

Feature	Must have	Should have	Could have	Not needed
Record / import purchases and stock	☑	☐	☐	☐
Demand forecast	☐	☑	☐	☐
Shortage / overstock risk	☑	☐	☐	☐
Price trends	☐	☐	☐	☑
Supplier price comparison	☑	☐	☐	☐
Purchase recommendation	☐	☐	☑	☐
Learning from my decisions	☐	☑	☐	☐
Budget utilization	☑	☐	☐	☐
Dashboard and alerts	☐	☐	☑	☐
Unusual transaction alerts	☐	☐	☐	☑
Chat assistant	☐	☐	☐	☑

Figure A.6: Client Requirement Questionnaire, page 6 of 7 (completed 14 October 2026)



J. Final questions

J1. How would you prefer to use the system? (choose one)

- ☑ Desktop / laptop browser
- ☐ Tablet
- ☑ Phone browser

J2. What would make you stop using such a system? (tick all)

- ☐ Recommendations I cannot understand
- ☐ Slow pages
- ☑ Hard data entry
- ☑ Wrong forecasts
- ☑ Privacy concerns
- ☐ Other: _____

J3. Anything else we should know?

Thank you for your time.

2026/10/15 02:18

Figure A.7: Client Requirement Questionnaire, page 7 of 7 (completed 14 October 2026)



Campus Store

Vendor Bill BILL/2026/10/0141

Invoice Date: 13/10/2026

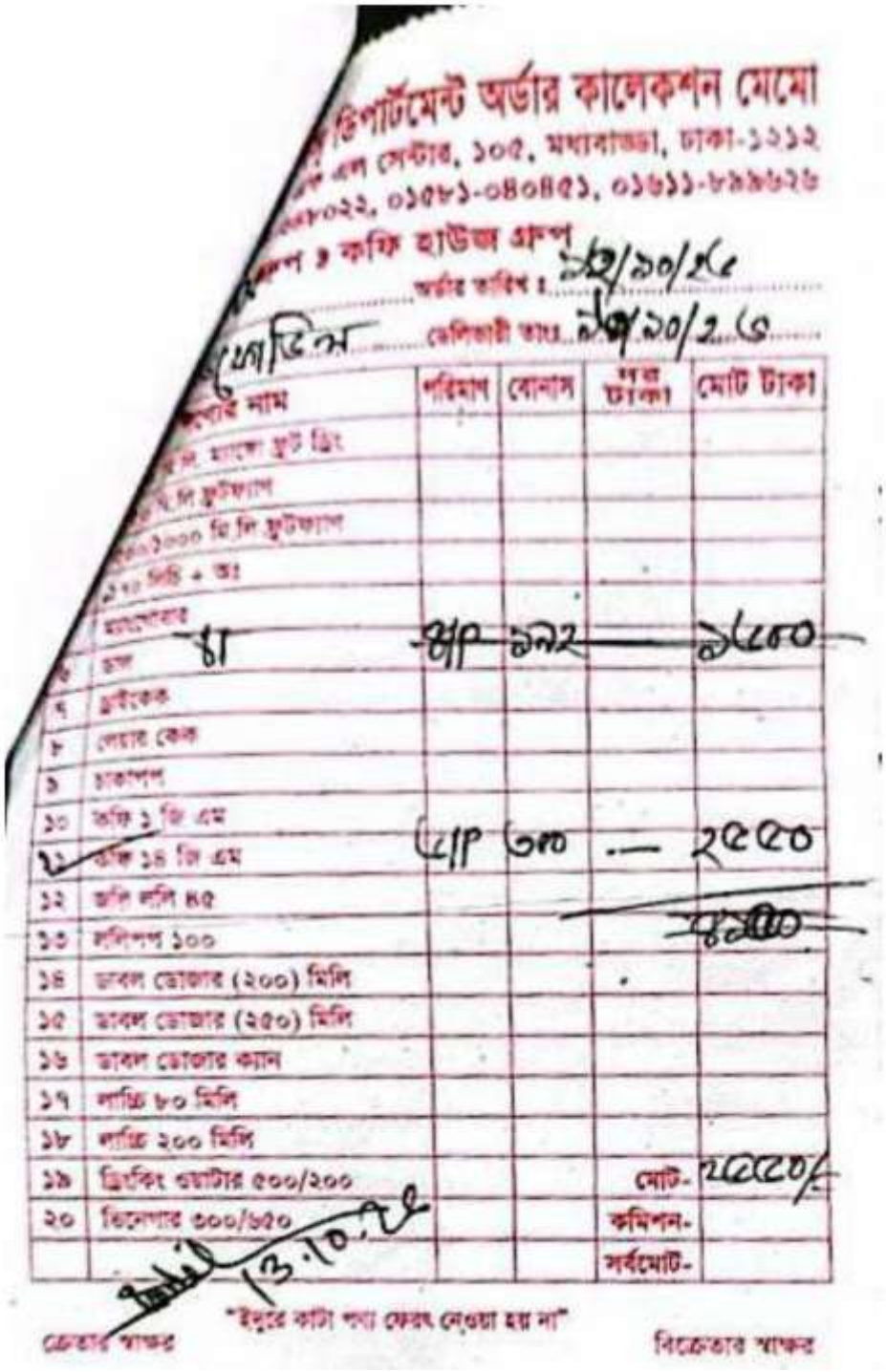
Source: P21303

Description	Quantity	Unit Price	Lot/Serial	Expiry Date	Taxes	Amount
P21303: Olympic Butter Bite Biscuits 15tk	24.00 Units	12.63				303.12
P21303: Olympic Nice Biscuit 50gm 10tk	24.00 Units	8.67				208.08
P21303: Olympic Elachi plus 10tk	24.00 Units	8.67				208.08
P21303: Olympic Wayfun Orange Biscuit 10 Tk	24.00 Units	8.58				205.92
P21303: Olympic Wayfun Pineapple Biscuit 10Tk	24.00 Units	8.58				205.92
P21303: Olympic Wayfun Chocolate Biscuit 10 Tk	24.00 Units	8.58				205.92
P21303: Olympic Elachi plus 190 gm	9.00 Units	42.00				378.00
P21303: Olympic Pulse Candy Masala 2Tk	400.00 Units	1.50				600.00
				Subtotal		2,315.04
				Total		2,315.00

Printed Date: 13-10-2026 17:51:09

Printed By: Smart City (Purch

Figure A.8: Campus supershop: printed vendor bill BILL/2026/10/0141 from Olympic Industries Ltd., 13 October 2026



ডিপার্টমেন্ট অর্ডার কালেকশন মেমো

এল সেক্টর, ১০৫, মধ্যবাড্ডা, ঢাকা-১২১২

০১৫৮১-০৪০৪৫১, ০১৬১১-৮৯৯৬২৬

গ্রুপ : কফি হাউজ গ্রুপ

অর্ডার তারিখ : ১৩/১০/২৫

ডেলিভারী তারিখ : ১৪/১০/২৬

ক্রমিক	পণ্যের নাম	পরিমাণ	বোনাস	প্রতি টাকা	মোট টাকা
১	ফ্লেভার্ড সিরাপ				
২	ফ্রুট ফ্লেভার্ড সিরাপ				
৩	১০০০ মি লি ফ্রুটফ্লেভার্ড				
৪	১২ ওয়াট + অন্য				
৫	ব্যাগেজ				
৬	চামচ	৪৮	১৯২	—	১৮০০
৭	ব্লেন্ডার				
৮	সেভার কেক				
৯	চাকাপদ				
১০	ডাবি ১ লি এম				
১১	ডাবি ১৪ লি এম	৫৮	৩০০	—	২৫৫০
১২	ডাবি লাবি ৪৫				
১৩	লাবিপদ ১০০				৪২০০
১৪	ডাবল ডোজার (২০০) মিলি				
১৫	ডাবল ডোজার (২৫০) মিলি				
১৬	ডাবল ডোজার ক্যান				
১৭	লাম্বি ৮০ মিলি				
১৮	লাম্বি ২০০ মিলি				
১৯	ব্রিকিং ওয়াটার ৫০০/২০০			মোট-	২৫৫০/-
২০	ডিসেপার ৩০০/৬২০			কমিশন-	
				সর্বমোট-	

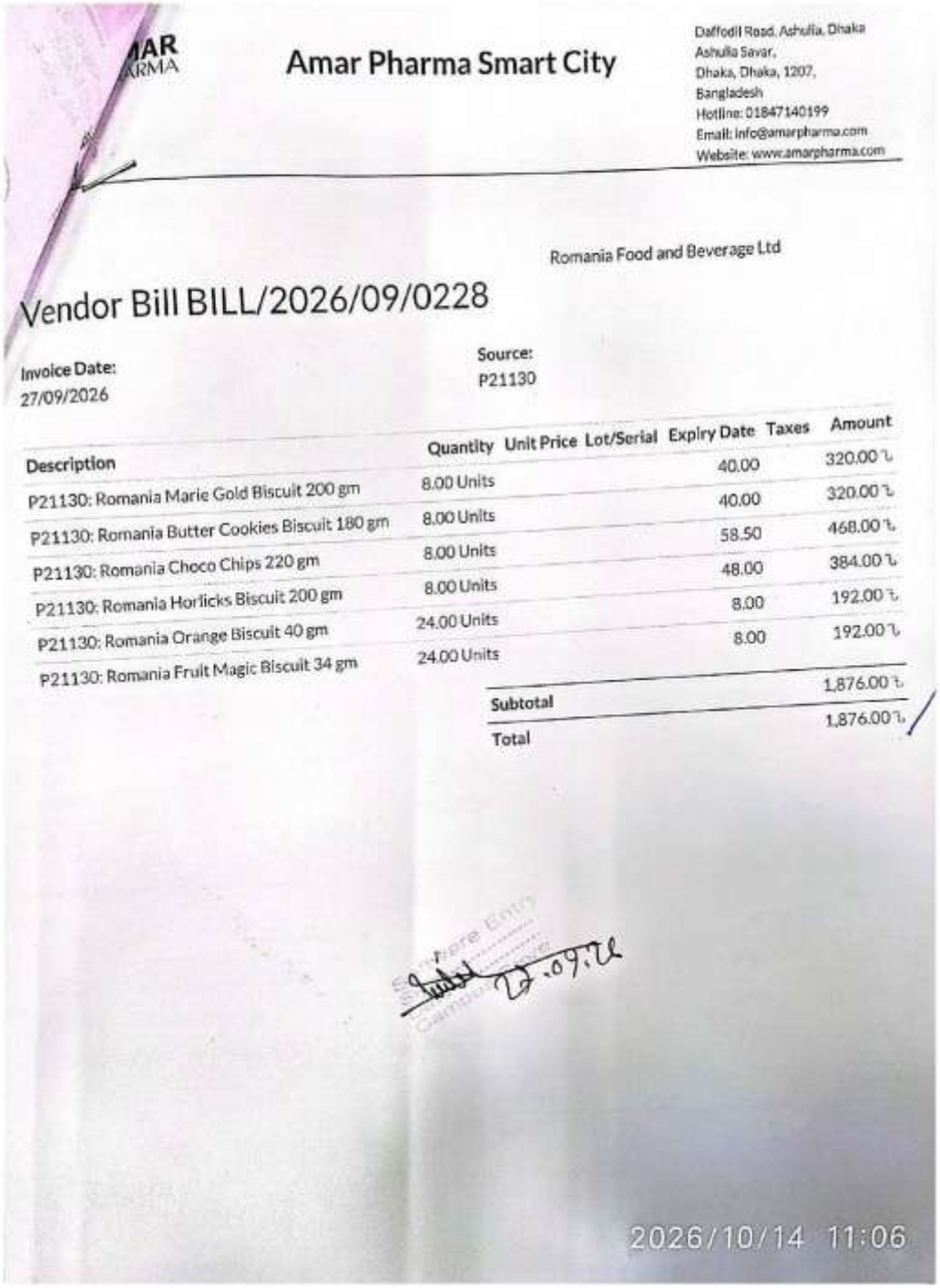
১৩.১০.২৫

"ইদুরে কাটা পণ্য ফেরৎ নেওয়া হয় না"

ক্রেতার স্বাক্ষর

বিক্রেতার স্বাক্ষর

Figure A.9: Campus supershop: handwritten supplier order collection memo (Bangla)



Amar Pharma Smart City

Daffodil Road, Ashulia, Dhaka
Ashulia Savar,
Dhaka, Dhaka, 1207,
Bangladesh
Hotline: 01847140199
Email: info@amarpharma.com
Website: www.amarpharma.com

Romania Food and Beverage Ltd

Vendor Bill BILL/2026/09/0228

Invoice Date:
27/09/2026

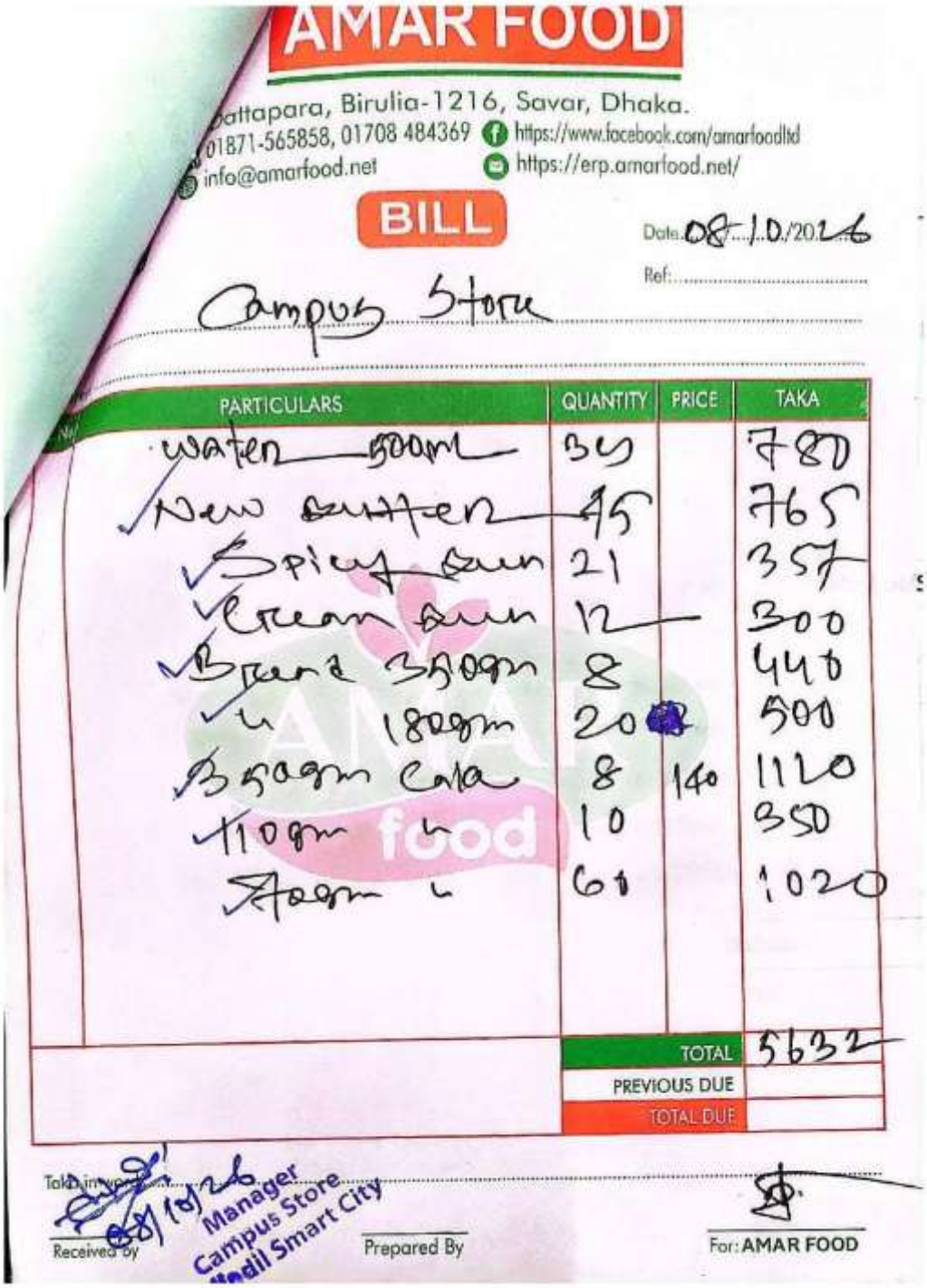
Source:
P21130

Description	Quantity	Unit Price	Lot/Serial	Expiry Date	Taxes	Amount
P21130: Romania Marie Gold Biscuit 200 gm	8.00 Units			40.00		320.00 ৳
P21130: Romania Butter Cookies Biscuit 180 gm	8.00 Units			40.00		320.00 ৳
P21130: Romania Choco Chips 220 gm	8.00 Units			58.50		468.00 ৳
P21130: Romania Horlicks Biscuit 200 gm	8.00 Units			48.00		384.00 ৳
P21130: Romania Orange Biscuit 40 gm	24.00 Units			8.00		192.00 ৳
P21130: Romania Fruit Magic Biscuit 34 gm	24.00 Units			8.00		192.00 ৳
				Subtotal		1,876.00 ৳
				Total		1,876.00 ৳

27.09.26

2026/10/14 11:06

Figure A.10: Campus supershop: printed vendor bill BILL/2026/09/0228 from Romania Food and Beverage Ltd., 27 September 2026


AMAR FOOD

Bhattapara, Birulia-1216, Savar, Dhaka.
01871-565858, 01708 484369 https://www.facebook.com/amarfoodltd
info@amarfood.net https://erp.amarfood.net/

BILL

Date: 08/10/2026

Ref:

Campus Store

PARTICULARS	QUANTITY	PRICE	TAKA
Water 500ml	39		780
New Sutter	45		765
Spicy Bun	21		357
Cream Bun	12		300
Bread 350gm	8		440
" 180gm	20		500
350gm Cola	8	140	1120
110gm	10		350
70gm "	61		1020
		TOTAL	5632
		PREVIOUS DUE	
		TOTAL DUE	

Taka in words: .....

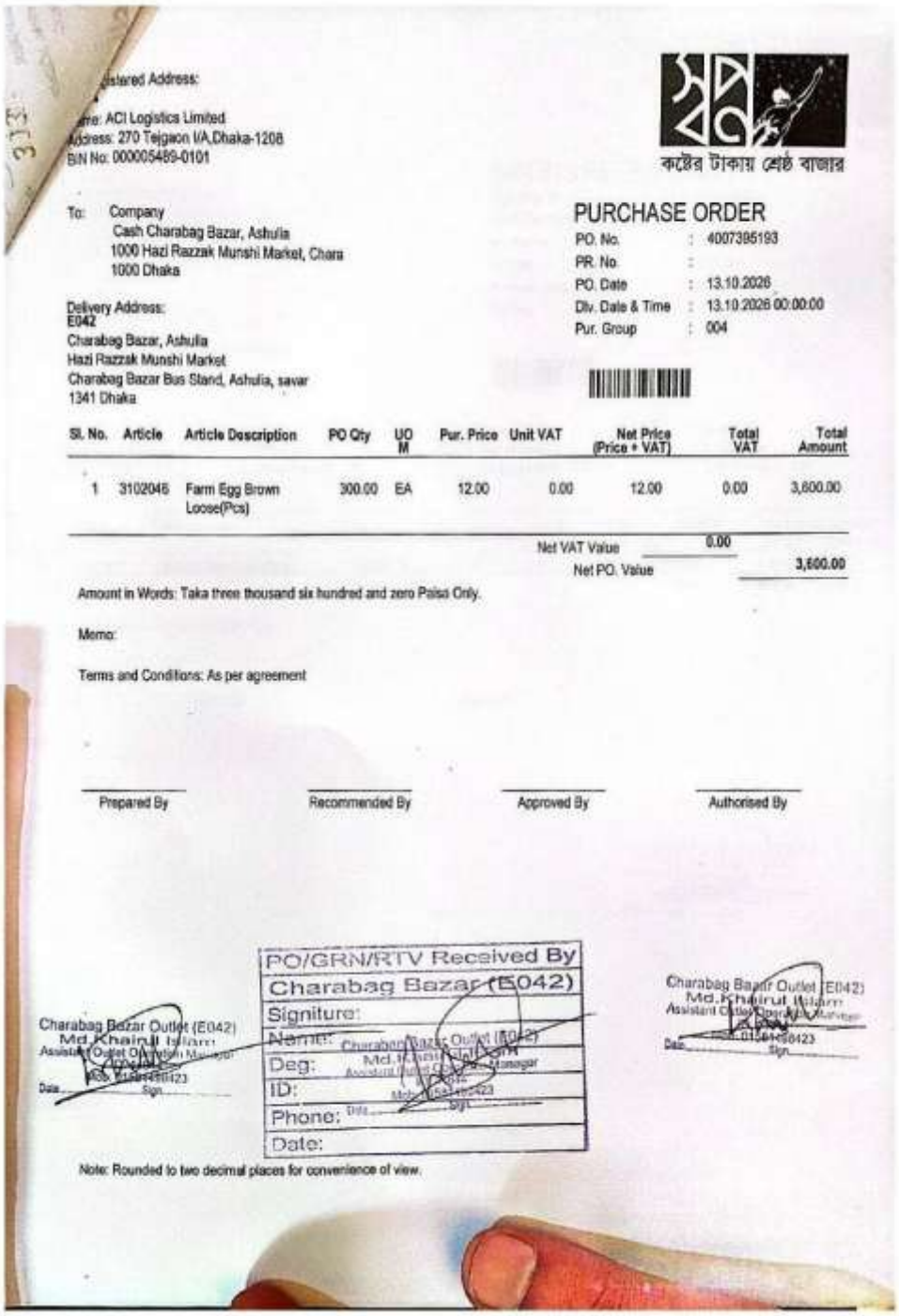
Received by

08/10/26
Manager
Campus Store
Modil Smart City

Prepared By

For: AMAR FOOD

Figure A.11: Campus supershop: handwritten bill from Amar Food, 8 October 2026



Registered Address:
Name: ACI Logistics Limited
Address: 270 Tejgaon I/A,Dhaka-1208
BIN No: 000005489-0101

কষ্টের টাকায় শ্রেষ্ঠ বাজার

To: Company
Cash Charabag Bazar, Ashulia
1000 Hazi Razzak Munshi Market, Chara
1000 Dhaka

Delivery Address:
E042
Charabag Bazar, Ashulia
Hazi Razzak Munshi Market
Charabag Bazar Bus Stand, Ashulia, savar
1341 Dhaka

PURCHASE ORDER

PO. No. : 4007395193
PR. No. :
PO. Date : 13.10.2026
Dlv. Date & Time : 13.10.2026 00:00:00
Pur. Group : 004

Sl. No.	Article	Article Description	PO Qty	UO M	Pur. Price	Unit VAT	Net Price (Price + VAT)	Total VAT	Total Amount
1	3102046	Farm Egg Brown Loose(Pcs)	300.00	EA	12.00	0.00	12.00	0.00	3,600.00

Net VAT Value 0.00
Net PO. Value 3,600.00

Amount in Words: Taka three thousand six hundred and zero Paisa Only.

Memo:

Terms and Conditions: As per agreement

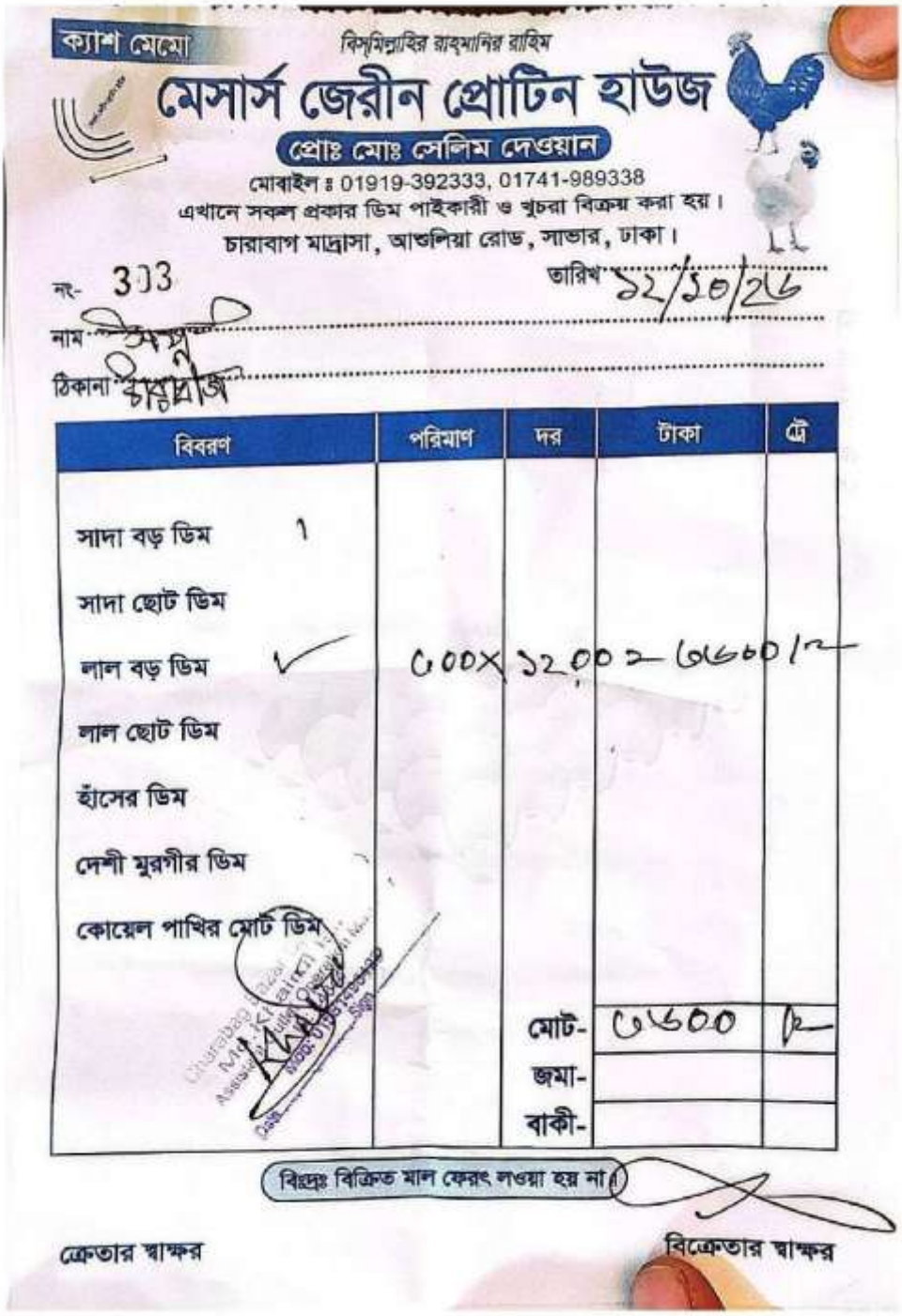
Prepared By | Recommended By | Approved By | Authorised By

PO/GRN/RTV Received By
Charabag Bazar (E042)
Signature:
Name:
Deg:
ID:
Phone:
Date:

Charabag Bazar Outlet (E042)
Md.Khairul Islam
Assistant Outlet Operation Manager
Mob: 01541480423
Date Sign

Note: Rounded to two decimal places for convenience of view.

Figure A.12: Swapno (Charabag Bazar outlet): purchase order 4007395193, 13 October 2026 (300 brown eggs)



ক্যাশ মেমো

বিসমিল্লাহির রাহ্‌মানির রাহিম

মেসার্স জেরীন প্রোটিন হাউজ

প্রোঃ মোঃ সেলিম দেওয়ান

মোবাইল ঃ 01919-392333, 01741-989338

এখানে সকল প্রকার ডিম পাইকারী ও খুচরা বিক্রয় করা হয়।

চারাবাগ মাদ্রাসা, আশুলিয়া রোড, সাভার, ঢাকা।

নং- 303

তারিখ 12/10/26

নাম স্বপ্ন

ঠিকানা চারাবাগ

বিবরণ	পরিমাণ	দর	টাকা	ট্রে
সাদা বড় ডিম	1			
সাদা ছোট ডিম				
লাল বড় ডিম	✓	600 × 12.00 =	6600/-	
লাল ছোট ডিম				
হাঁসের ডিম				
দেশী মুরগীর ডিম				
কোয়েল পাখির মোট ডিম				
		মোট-	6600	/-
		জমা-		
		বাকী-		

বিঃদ্রঃ বিক্রিত মাল ফেরৎ লওয়া হয় না।

ক্রেতার স্বাক্ষর

বিক্রেতার স্বাক্ষর

Figure A.13: Swapno (Charabag Bazar outlet): handwritten supplier cash memo for eggs (Bangla)